Abdelmouhaimen Sarhane

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Skills

Skills: Python, C++, PyTorch (Lightning), Qualcomm AIMET, OpenCV, SQL, ROS2, OpenGL, MLFlow, Flask, FastAPI, GCP, BigQuery, PowerBI
Soft Skills: Problem-solving, Curiosity, Adaptability, Communication, Collaboration
Languages: French (Bilingual), English (Bilingual), Arabic (Native), Spanish (Beginner)

Experience

Parrot Drones — Paris, France
Computer Vision & Embedded Deep Learning Engineer — *September 2025 – Present*

- Working on **High Speed** & 360° **Obstacle Avoidance** *perception/guidance*.
- Designed and deployed a **real-time object filtering and tracking pipeline** for the **Anafi UKR** drone, optimizing **latency** and **robustness** under onboard constraints, from **Python prototype** to **production C++**.
- Built **Object detection pipelines** and improved **tracking accuracy** and downstream system reliability.
- Developed a **deep learning stereo depth estimation solution** for **next-generation drone platforms**, covering **model architecture**, **dataset preparation**, **evaluation**, and **embedded integration**.
- Applied **quantization** and **model optimization** on **Qualcomm hardware** using **AIMET**, reducing inference time by **70%** and achieving around **100 FPS**.
- Created **MLOps** and **offline evaluation pipelines** to accelerate model iteration, experimentation, and benchmarking.
- Conducted **field testing** and **debugging**, diagnosing perception issues related to **camera inclination**, **camera decalibration**, **fisheye stereo geometry**, and **propeller interference**, while investigating and validating solutions to improve system robustness.
- Contributed to the drone software stack across **build**, **deployment**, and **embedded SDK workflows**, collaborating with **deep learning**, **stereo sensing**, and **embedded systems** teams.

Renault Group (Ampere Software Technology) — Toulouse, France
AI Research - Semantic Scene Perception — End of studies Internship — *March 2025 – July 2025*

- Contributed to the development of an AI-based driving assistant system for scene understanding and risk detection.
- Adapted and fine-tuned pretrained models on driving-specific datasets, achieving a 94% accuracy in classification and object detection
- Benchmarked perception model performance and investigated state-of-the-art fine-tuning strategies for foundation models.
- Authoring an internal report on finetuning models on behavior-aware scene understanding for intelligent vehicles.

IRIT Github — Toulouse, France
Computer Vision & Embedded Systems Developer — *Oct 2024 – Mar 2025*

- Designed a system for adaptive image sampling and reconstruction using compressed sensing and deep learning.
- Implemented two networks: NetM for mask generation and NetE for inpainting-based reconstruction.
- Optimized bandwidth usage in drone-based transmissions; added real-time refinement loop.
- Developed a Streamlit dashboard for interactive monitoring and visualization.

GET-OMP (CNRS) Github — Toulouse, France
Deep Learning Intern - Satellite Image Processing — *Jun 2024 – Sep 2024*

- Developed U-Net with attention to detect water bodies in West Africa from Landsat satellite imagery.
- Achieved 94% F1-score through preprocessing and prediction optimizations.
- Performed temporal analysis from 1984-present for hydrological insights.
- Integrated NDWI and MNDWI indices in geospatial pipelines with GDAL and QGIS.

ENAC — Toulouse, France
HPC and Data Analysis Intern *Paper Link* — *Jun 2023 – Jul 2023*

- Assessed energy consumption of matrix computations on compressed data & Co-authored a paper accepted at ICT4S 2024.
- Built efficient C++ workflows for scientific computing with energy monitoring. (Blaz compressor, PowerJoular)
- Applied advanced compression to sports data heatmaps, reducing compute overhead by 25%.

Education

ENS Paris-Saclay (Master MVA – Mathematics, Vision, Learning) | Paris, FR
MScR Master of Research — Specialized in Artificial Intelligence and Computer Vision
French highly selective and widely recognized ***M2 research program*** *in* ***artificial intelligence, machine learning, and computer vision****, regarded as a leading pathway into top-tier AI research and industry roles.*

INP-ENSEEIHT | Toulouse, FR — *September 2022 – October 2025*
MEng Master of Engineering — Digital Sciences "Diplôme d'ingénieur": Image and Multimedia GPA: 3.92/4.0
Leading French engineering program in ***digital sciences****, with strong training in* ***AI, machine learning, software engineering, embedded systems, cloud computing, and high-performance computing****.*

Projects

Manysight | AI Camera Intelligence & Spatial-Awareness Platform ManySight.com GitHub
- Built an **SDK and AI platform for existing camera systems**, turning live camera streams into structured, queryable scene understanding for **multimodal AI assistants**.
- Developed **multi-camera spatial awareness** using calibrated cameras, object detection/tracking, and geometric reasoning so agents can reason about where people and objects are across a physical space.
- Designed the perception-to-agent pipeline for **natural-language scene queries, event extraction, and real-time visual analytics**, with an emphasis on reusable APIs and deployment-ready computer vision components.

Sotiby | AI Voice Agent & Business Automation Platform Sotiby.com *Demo on demand*
- Built an **end-to-end voice AI agent platform** with phone-number integration for natural, real-time customer interactions.
- Integrated **appointment scheduling, ERP/business tools, and API-based actions**, enabling the agent to complete operational workflows rather than only answer questions.
- Implemented configurable agent workflows and conversation orchestration to support multiple business use cases from a single platform.

Kvesty (Daregram) | Real-Time Social Challenge Platform *kvesty.com*
- Designed and built a **full-stack TypeScript** a **30-screen React Native/Expo** mobile app, a **Next.js** companion client, and a **GraphQL/Prisma** backend supporting challenges, communities, messaging, moderation, and proof-based participation.
- Engineered **end-to-end platform infrastructure**: **Clerk** auth, **Cloudinary** signed media uploads, Upstash **Redis** rate limiting, **Expo** push notifications, and **Resend** emails, over a **27-model** PostgreSQL **schema** with authorization, pagination, and admin moderation workflows.
- Implemented **real-time chat, map-based discovery, and geolocation-verified submissions** using WebSocket subscriptions and Apollo; optimized media-heavy feeds with Apollo cache policies, pagination, and FlatList virtualization.
- *Technologies Used: TypeScript, React Native, Expo, Next.js, GraphQL, Prisma, PostgreSQL, Clerk, Cloudinary, Upstash Redis, WebSockets, EAS*

Polymer AI Formulator — Junior Entreprise Mission *(TeknorApex X OSS Ventures)*
- Audited and structured internal formulation data (raw materials, properties, test results) into a standardized data model with typed metadata and quality controls.
- Performed domain modeling via PCA, clustering, and correlation analysis to map input/output relationships and visualize the formulation design space.
- Evaluated technical approaches (generative AI, constraint optimization, hybrid ML/physics) and delivered a feasibility and interpretability assessment with a target model recommendation.
- Defined the technical roadmap: data pipeline architecture, KPIs, success metrics

AI-Powered Clinical Documentation & Patient Chatbot Platform *cabinetplus.ma*
- Automated transcription of medical visits into structured reports, including prescription summaries and care advice
- Implemented patient-specific chatbot using RAG over all patient documents (scans, history, prescriptions)
- *Technologies Used: Langchain,Llamaindex, LRAG, Docker, PostgreSQL, React, Typescript*

Multimodal Data Processing Projects *(ENSEEIHT)*
- Image stitching: SIFT, homography estimation, KLT tracking — Source separation: NMF, U-Net audio models
- Segmentation: K-means, SLIC superpixels — 3D streaming: compression, remeshing, camera pose estimation, 3D rendering
- 3D Inverse Problems: SfM, MVS, Shape-from-shading, stereo photometry.

3D Projector Pose Adjustment | *GitHub*
- Developed a computer vision system to align prehistoric paintings onto a replica of the Chauvet Cave.
- Used Structure-from-Motion (SfM) and Multi-View Stereo (MVS) to reconstruct a 3D model of the replica.
- Calibrated projectors using Zhang's method (treated as inverted cameras) and applied ICP for alignment.
- Enhanced projection accuracy and minimized manual positioning errors.

Additional Projects: 3D Rendering Engine (Java), 3D Surface Compression and Remeshing using MAPS (Python), Traffic Sign Recognition using Deep Learning (Python, TensorFlow), Simplified Language Compiler (OCaml), Learning app: RevEasy (Java, Swing), RAG-based Document Chatbot